

INSTRUCTOR OPERATIONS / COURSE 1

Sessions 03-05 Instructor Manual

Data evidence, a working app, and a controlled revenue system

03 DATA EVIDENCE	04 WORKING APP	05 REVENUE SYSTEM	120 MIN EACH
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Version 1.0 | Prepared 30 July 2026 | Instructor and evaluator use only
Authored packages; validation and rehearsal pending

This is the compact operator reference for delivering Sessions 3, 4, and 5. It consolidates the authored source packages into one classroom sequence without promoting any package beyond its recorded lifecycle.

Truth boundary. All three source packages are **Authored**. None is yet Validated, Loaded, Deployed, Piloted, or Rehearsed. A source ledger marked "Verified: 30 July 2026" means the cited claim was checked on that date; it does not mean the session, tool path, LMS path, grader, or fallback has passed the course validation gate.

1. Operator orientation

1.1 The three-session arc

Session	Transferable capability	Student-owned proof	Public boundary
03 - Working with data, using AI	Frame a bounded query, execute where the data lives, and verify independently	Verified data memo, scale trace, two-method check, visual judgment	Method sentence only; no TrustMRR row or derived value
04 - A proven pattern to a working app	Convert public evidence into an original, testable feature contract and ship a verified V1	Product/prompt checkpoint, public app, AT log, V1/V2 lineage	Safe public app and approved gallery card; no copied brand, private data, prompt log, or grade
05 - Revenue systems with Make	Design a controlled operating system with explicit state, recovery, approval, and evidence	Revised flowchart, blueprint, five-fixture trace, sample output, workflow PNG	Privacy-cleared copy/blueprint and sample output; no credentials, PII, grades, or unsafe live action

1.2 Room roles

- **Lead instructor:** pace, reveal boundaries, teach the decision model, and protect assessed work time.
- **Teaching assistant:** recover access and execution without making the learner's decision or completing the analysis/build.
- **LMS operator:** test-student check, section gates, exception logging, autosave/submission monitoring, and incident record.
- **Evaluator/finaliser:** inspect evidence, apply deterministic policy, review provisional AI output, and record any override reason.

For Session 3, use two TAs when possible: one for the formula/pivot lane and one for Colab. If one instructor is alone, run the guided proof on the formula lane and keep Colab as an extension/follow-up. Tool choice never changes the score ceiling.

1.3 Evidence and lifecycle legend

Label	Meaning in this manual	Operator action
SOURCE	Timing, gate, rule, or checklist is stated in the named session package	Preserve exactly unless a versioned correction is approved
MANUAL CONTROL	A consolidated operating check derived from the packages but not separately authored at that timestamp	Use as a safety hold; do not describe it as package validation
Authored	Content and contracts exist	Recheck release dependencies before delivery
Source-verified	A volatile claim was checked on the package date	Recheck at T-7; log reviewer and date
Validated	Course release dependencies and target LMS/tool paths have passed	None of Sessions 3-5 currently carries this state

1.4 Stop-the-line conditions

Do not expose a private key, expected result, hidden exemplar, real credential, private data row, or evaluator prompt to keep the room moving. Pause or invoke the named fallback when any of these conditions occurs:

- a dataset version or checksum does not match;
- a public app or scenario contains real/private data, an apparent secret, or an unlabelled external integration;
- a risky action can send, publish, delete, pay, or change production state without recorded approval;

- a fallback would remove the required evidence or lower the grade ceiling;
- an AI result is being treated as a final grade without instructor finalisation;
- a vendor/tool claim failed its T-7 recheck and affects learner instructions, grading, or a quiz answer.

2. Master preflight

2.1 T-7 release hold - SOURCE

Session 03

- ☐ Recompute the private objective key from the frozen source; reconcile spreadsheet and pandas outputs.
- ☐ Confirm dataset version and learner-file SHA-256; keep the answer pack private.
- ☐ Confirm the large derivative has reproducible token-count proof above 1,000,000 tokens and is labelled derived/synthetic where applicable.
- ☐ Inspect representative-sample edge coverage without making a graded answer inferable.
- ☐ Run grader calibration fixtures, including prompt injection and low-confidence cases.
- ☐ Recheck Google Sheets functions, Colab behavior, and pandas CSV APIs.
- ☐ Clean-runtime test the learner Colab and local fallback; confirm the offline pack per section.
- ☐ Confirm six visualization items and text/table equivalents are present and section-equivalent.

Session 04

- ☐ Record current public Liinks evidence and round volatile revenue in teaching copy.
- ☐ Recheck the founder evidence plus Liinks block, analytics, plan, API, and domain pages.
- ☐ Recheck Lovable Free daily/monthly credits, Plan-mode behavior/cost, Build variability, and Publish-dialog behavior.
- ☐ Run the golden prompt in a fresh Free workspace and log observed credits without turning the observation into a promise.
- ☐ Exercise both Plan branches: formal editable plan and `NO_FORMAL_PLAN` recovery.
- ☐ Confirm the fresh-Free golden run and mixed-confidence dry-run thresholds still pass before calling the 96/100 score validated.
- ☐ Replace or remove stale quiz distractors affected by vendor changes.
- ☐ Validate the fallback starter/restore path and its ownership/credit behavior in the current UI.

Session 05

- ☐ Reverify every Make claim in the source ledger against official documentation.
- ☐ Open the public scenario page logged out; inspect all visible settings, mapped values, notes, author identity, and dependency warnings.
- ☐ Import the blueprint into a blank scenario; confirm JSON parses, size is below 2 MB, and connections did not travel.
- ☐ Run normal, duplicate, malformed, timeout, and approval fixtures against the matching expected-results manifest.
- ☐ Inspect all static values, mapped samples, prompts, notes, URLs, filenames, screenshots, and outputs for secrets/PII.
- ☐ Capture a fresh redacted run log and sample output.
- ☐ Verify gates S5.1-S5.4 with a test student and load-test the formative feedback/bypass path where scheduled.
- ☐ Cache deck, templates, fixtures, corrected reveal, blueprint, and outage replay locally; keep scheduling off and use demo-only connections.

2.2 T-24 delivery hold

Timing provenance: Session 04 defines T-24 explicitly. Applying the same hold to Sessions 03 and 05 is a MANUAL CONTROL for a consolidated operator workflow.

- ☐ Freeze section, roster, room, lead, TAs, LMS operator, and escalation contact.
- ☐ Record the source-recheck reviewer/date and any versioned copy changes.
- ☐ Confirm student-visible materials contain no instructor answer, private filename, hidden exemplar, evaluator prompt, or expected-results manifest.
- ☐ Complete a test-student path through every gate, draft/autosave field, file constraint, final submit, and recovery exception.
- ☐ Confirm the offline/accessibility pack: tagged PDF or HTML, plain text, captions/transcript, text equivalents, and keyboard route.
- ☐ Confirm demo accounts contain only fictional/synthetic data and no saved secrets, personal history, or production connections.
- ☐ Confirm the grader and deterministic-check jobs are configured but remain provisional; name the human finaliser.
- ☐ Record the published deadline and exception rule for the section.

Session 04 source-defined additions:

- ☐ Run the first prompt in a fresh Lovable Free workspace; record cost and reset the demo project.

- ☐ Confirm the Liinks golden-run gate; if it has not passed, present the score as provisional.
- ☐ Preflight daily-exhaustion versus monthly-cap exceptions without collecting billing screenshots.
- ☐ Confirm checkpoint, V1, V2, and immutable-history behavior.
- ☐ Open public TrustMRR and official feature/tool pages; never project the private class Sheet.
- ☐ Prepare the tested SignalShelf starter, captioned recording, static checkpoint pack, 60-minute timer, and AT-01-AT-18 cards.
- ☐ Verify back-of-room legibility at 1440 x 900 and remove all real emails, tokens, keys, and private rows from the demo.

2.3 T-30 room preflight

Session 03 - SOURCE

- ☐ Log in as the section's test student.
- ☐ Confirm Session 3 is open while the large-scale pack and visual check remain closed.
- ☐ Download the small dataset and compare SHA-256 with the private manifest.
- ☐ Type, refresh, and confirm assessment autosave.
- ☐ Open the Colab copy link in an incognito profile and run all cells.
- ☐ Confirm projector/browser history/notes/private filenames cannot reveal instructor material.
- ☐ Put the fallback pack on the instructor machine and one encrypted USB or managed-Drive location.

Session 04 - SOURCE

- ☐ Repeat the T-24 environment checklist.
- ☐ Verify the checkpoint unlocks V1 and the strong prompt/fallback remain hidden before the checkpoint.
- ☐ Confirm Publish uses the dialog and the incognito/public-route checks are ready.
- ☐ Display the 60-minute ship clock; assign the credit/outage exception operator.

Session 05 - MANUAL CONTROL derived from the environment and room setup

- ☐ Run the normal fixture once in the demo scenario and reset the audit store before learners arrive.
- ☐ Confirm every learner can access one initial template, one failure table, and the correct workflow pack.
- ☐ Verify the expected-results manifests and corrected flowchart remain instructor-only before Gate 1.

- ☐ Confirm each pair can work while retaining individual scenario, blueprint, trace, and submission ownership.
- ☐ Project the execution trace only; hide connection dialogs, account chrome, URLs, and personal identity.
- ☐ Keep the 35-minute outage replay, module/state cards, and blank trace table ready.

2.4 Incident record

For every incident, record: section, timestamp, affected material/gate/service, number affected, fallback invoked, duration, assessment impact, recovery deadline, and remediation. Do not record private answer values, dataset rows, credentials, billing screenshots, or sensitive learner/account details.

3. Session 03 - Working with data, using AI

3.1 Promise, ownership, and boundaries

Student proof: one individual verified data memo. The objective fields bind to a private checksum/version-controlled answer pack. Judgment and visual rationales may receive provisional AI feedback; the instructor finalises the artifact grade.

Non-negotiables:

- individual graded answers; method discussion only after the first checkpoint;
- tool choice never affects score;
- do not confirm values during the timed challenge;
- never show TrustMRR rows or private aggregates on a public/shared surface;
- schema/sample informs method but cannot prove the full-file result;
- the teaching slice is stratified and edge-preserving, not population-representative;
- the second method must differ mechanically, not repeat the same model/prompt/assumption.

3.2 Exact 120-minute run of show

Time	Min	Mode	Instructor move	Learner work / evidence
00:00-00:05	5	Retrieval	Revisit "Sheets or upload"; collect three failure predictions.	Write one failure condition; pair-share.
00:05-00:11	6	Consequential hook	Show two confident outputs using different denominators; reveal no answer.	Vote and name the hidden choice.

Time	Min	Mode	Instructor move	Learner work / evidence
00:11-00:16	5	Contract	State privacy, individual-work rule, version, unit, and "AI answer = hypothesis until checked."	Open dataset/assessment; record version ID.
00:16-00:38	22	Independent challenge 1	Release S3-DATA-01-09; use recovery questions only.	Complete six objective and three judgment items; autosave.
00:38-00:46	8	Checkpoint / repair	Snapshot for two minutes; reveal denominator traps, not answers.	Repair one answer; label observation, inference, limitation.
00:46-00:54	8	Live proof: scale wall	Show filename/version, byte/token method, and checksum; explain fit is neither guaranteed nor sufficient.	Predict next route and trade-off.
00:54-01:00	6	Decision ladder	Teach inspect -> ask -> formula -> schema/sample -> code -> verify/visualize.	Complete S3-SCALE-01.
01:00-01:22	22	Guided build	Supported lane: AI-written Sheets formula. Core lane: schema/sample to pandas in Colab.	Produce S3-SCALE-02/03 and capture executable working/output.
01:22-01:31	9	Verification / repair	Reconcile base rows -> valid rows -> grouped totals -> independent recomputation.	Complete S3-DATA-10; report and explain the gap.
01:31-01:39	8	Visual judgment	Teach compare, distribute, relate, change, compose; show axis and missingness traps.	Diagnose why the prettier chart is weaker.
01:39-01:54	15	Scenario check	Open S3-VIZ items; at minute 10 remind learners to name decision and encoding.	Select/defend six choices; submit formative check.
01:54-01:58	4	Ship	Show required fields and privacy boundary.	Submit memo, answer set, scale trace, and verification evidence.
01:58-02:00	2	Bridge	Connect verified data evidence to a product contract.	Save a public-safe method sentence.

Protected student work: 72 minutes. If the room is behind, compress instructor discussion before the challenge, guided build, verification trace, or scenario check. At challenge minute 11, use draft completion only: if fewer than 70% have reached S3-DATA-04, add three challenge minutes and remove them from whole-room discussion, not verification.

3.3 Facilitation moves

When asked "Is this right?", use one recovery prompt:

- "Show me the row grain and denominator."
- "Which rows did you exclude?"
- "What does blank mean here?"
- "Where is the unit?"

- "Can the working be rerun after the file changes?"
- "What is one independent check?"

Formula-lane success is an explicit query contract, readable formula, execution in Sheets, independent pivot/formula check, and recorded gap. Colab-lane success is schema/sample to code, required-column/null assertions, execution against the gated full file, row counts before aggregate, compact output, and no public notebook containing private outputs.

3.4 Gates and submission timing

Time / event	Gate or surface	Operator action
T-30	Session page/open materials	Test-student exception only; verify flow.
00:00	Page, first materials, data-memo assignment, mixed data assessment	Open; draft/autosave on; correctness hidden.
00:38	Challenge checkpoint	Snapshot completion; do not close the assignment.
00:46	Scale manifest, schema, sample, gated data	Open for this section only.
01:39	Visualization check	Open one in-class attempt.
01:54	Visualization check	Close; feedback may release.
01:58	Data memo	Remind; keep open until the published section deadline.
Published deadline	Assignment + mixed assessment	Close; preserve queued drafts and submission state.

Final submit creates immutable Version 1. One instructor-authorized Version 2 correction window may open after provisional feedback or a documented incident; both versions remain visible. A dataset/key defect triggers a cohort-wide regrade and notification, never a learner penalty.

3.5 Grading and review flow

1. Verify dataset version and SHA-256 against the private manifest.
2. Run deterministic validators for objective items, internal consistency, and the two-method arithmetic.
3. Run static checks on formula/code: known columns, null/zero policy, output contract, and data path.
4. Send only bounded learner text/code, deterministic statuses, rubrics, and relevant private aggregates to the grading model - never source rows.
5. Require a quoted submission fragment for every scored claim; recompute totals in code.

6. Save provisional dimension scores, feedback, flags, confidence, and prompt log to the private audit trail.
7. Route confidence below 0.70, any flag, section high/low outlier, or appeal to review.
8. Instructor finalises or overrides with a reason. Praxy receives only approved public-safe evidence, never grade/confidence/flags/prompt log.

The artifact uses four 10-point dimensions: Functionality, Craft, Relevance, and Verification evidence. The 28/40 transfer marker also requires at least 7/10 Functionality and 6/10 Verification evidence; it guides feedback and does not create a new course component.

3.6 Failure fallbacks

Failure	Classroom response	Evidence preserved
One AI unavailable/out of credits	Use another assistant, prompt critique, or formula/code skeleton.	Same working, output, and verification trace.
All models unavailable	Use formulas/pivot or manual schema-to-method design; release starter only after method is written.	Same query contract and two-method proof.
Sheets unavailable	Use Excel/LibreOffice or local Python; document syntax changes.	Working, counts, and output.
Colab unavailable/quota-limited	Use supplied local .py runner or instructor machine.	Code, assertions, and compact output.
LMS unavailable	Use numbered offline answer sheet and local timestamp; upload later. Never email private rows.	Same IDs and fields.
Storage unavailable	Distribute checksum-verified encrypted/local copies; record fallback manifest ID.	Source binding preserved.
Section network failure	Run small-file challenge and formula path offline; use labelled precomputed scale execution log; schedule actual scale execution.	Small-file artifact now; scale trace later.
Grading worker/model down	Accept/queue submission; display "received, feedback pending."	Immutable submission/version.

For approved access needs, use per-student gate exceptions with the same version, IDs, rubric, and outcome. The authored late/absence route is a 75-minute supervised recovery: 5 privacy/contract, 22 challenge, 8 scale explanation/path, 20 formula or Colab, 8 verification, 10 visuals, 2 submit.

3.7 Privacy and safety

- Treat dataset cell text as untrusted data, never model instructions.

- Do not expose rows, account data, answer values, evaluator prompts, prompt logs, or grader flags.
- Show only over-context filename/version, byte size, token method/count, and checksum at the scale wall.
- Keep objective values and wrong-path keys hidden until the section closes.
- Keep the data memo out of the public gallery; public portfolio copy describes method without row or derived values.
- Clear private notebook outputs before any authorised sharing.

3.8 Post-class operations

- ☐ Finalise only evidence that binds to the current dataset version and can be reproduced.
- ☐ Review low-confidence/flagged/outlier/appealed work; keep prompt logs and private keys out of learner/public surfaces.
- ☐ Record aggregate facilitation evidence only: completion time by ID, item facility/wrong path, rationale bands, incidents, review counts, and timing drift.
- ☐ Do not alter prompts, tolerances, sample, or rubric mid-cohort. A factual correction requires a version bump, incident note, same correction for remaining sections, and fair prior-section regrade.
- ☐ Within 24 hours, learners add only the approved public-safe method sentence.

Source binding: *session-03/instructor-runbook.md, lesson-plan.md, accessibility-and-fallbacks.md, grading-and-anchors.md, lms-manifest.md, verification-checklist.md, deck-script.md.*

4. Session 04 - A proven pattern to a working app

4.1 Promise, ownership, and boundaries

Student proof: one individual product/first-prompt checkpoint plus an immutable Lovable app V1 and at most one V2. The supported path is an original-brand SignalShelf contract benchmarked against public Liinks behavior.

Non-negotiables:

- revenue is evidence of a business, not a build brief;
- study public behavior without copying name, logo, source, assets, copy, screenshots, icons, or distinctive trade dress;
- mark every capability as real, mock, or out of scope;
- no paid plan or credit purchase is required;
- a public URL is not proof until incognito, negative, keyboard, mobile, and evidence checks run;

- the strong prompt and conditional fallback remain hidden until the learner submits the checkpoint.

4.2 Exact 120-minute run of show

Time	Min	Mode	Instructor move	Learner work / evidence
00:00-00:04	4	Arrival	Show credit/privacy preflight; open Lovable, LMS, handout.	Confirm access; put away real data.
00:04-00:10	6	Retrieval prediction	Ask highest MRR vs easiest UI vs strongest one-hour proof.	Choose and write one reason.
00:10-00:17	7	Consequential hook	Show rounded current revenue for three candidates; reveal scorecard.	Re-score for feasibility/integration risk.
00:17-00:24	7	Concept	Contrast research, feature contract, copying, and parity; show mock/non-affiliation rules.	Mark candidate features real/mock/out.
00:24-00:32	8	Independent checkpoint	Release product + prompt checkpoint; use checklist only.	Submit product, sources, rationale, transfer, contract, first prompt.
00:32-00:35	3	Reveal	Explain vague clone failure; reveal instructor prompt structure.	Repair one missing prompt element.
00:35-00:40	5	Ship 00-05	Send one Plan message; explain no code edit and current one-credit behavior.	Adapt bounded Plan prompt; mark <code>NO_FORMAL_PLAN</code> if needed.
00:40-00:45	5	Ship 05-10	Review truth, AT coverage, state, failures, access, order; release fallback after checkpoint if needed.	Edit plan, record two owned edits, approve/proceed once.
00:45-00:53	8	Ship 10-18	Demonstrate shell, routes, seed, public/share state.	Build shell/seed; run AT-01/02.
00:53-01:07	14	Ship 18-32	Build identity, socials, and block operations.	Add/edit/order/hide/delete six types; AT-03-08.
01:07-01:17	10	Ship 32-42	Add original design controls/live preview; prohibit pixel copying.	Verify theme/readability; AT-09.
01:17-01:24	7	Ship 42-49	Add persistence, share-state encoding, and input failures.	Refresh/share/negative tests; AT-10-12.
01:24-01:30	6	Ship 49-55	Add labelled browser-local analytics; distinguish build agent vs in-app AI.	Generate/reset demo events; AT-13-15.
01:30-01:35	5	Ship 55-60	Publish through dialog; show security/public warning.	Publish, copy URL, begin AT-16.
01:35-01:46	11	Peer verification	Pair across teams; incognito, mobile width, keyboard before design opinion.	Return PASS/FAIL evidence; AT-16-18 and initials.

Time	Min	Mode	Instructor move	Learner work / evidence
01:46-01:52	6	Repair/evidence	Model an AT-specific repair; stop broad redesign.	Repair one stop-line defect or record limitation.
01:52-01:56	4	Submit V1	Show required fields and lock behavior.	Submit URL, log, evidence, limitation, consent.
01:56-01:59	3	V2 control	Explain monthly cap, no daily rollover, latest-eligible grading, 10-day window.	Select one failed test and one craft/access V2 item.
01:59-02:00	1	Bridge	Connect app events to a revenue-supporting workflow.	Name one event to automate in Session 5.

Protected student work: at least 84 minutes. Compress scorecard/debrief before the 60-minute ship clock or verification.

4.3 Plan and build control

Use the three-line narration cadence:

1. "The contract says..."
2. "The current build proves/fails it because..."
3. "The smallest next change is..."

If Plan mode asks a question or produces no formal editable plan, save the response, record `NO_FORMAL_PLAN`, and do not spend a second exploratory Plan credit. After checkpoint submission, release the fallback. The learner replaces bracketed decisions, records at least two owned edits, and switches once to Build mode with the prescribed implementation preface. If that current-UI path was not T-7-tested, use the tested starter and record `tool-access`; do not improvise a new live workaround.

4.4 Gates and submission timing

Time / event	Gate or surface	Operator action
00:00	Product/first-prompt checkpoint	Open. Formative AI feedback cannot block the build gate.
00:17	Acceptance contract	Open student contract/tests.
00:32	Checkpoint due	Submission unlocks V1; reveal strong prompt after checkpoint window.
After checkpoint, conditional	Fallback plan	Instructor releases only for <code>NO_FORMAL_PLAN</code> .
01:52	V1 submission prompt	Collect required evidence; app gate was already unlocked by checkpoint.
02:00 / section V1 deadline	Version 1	Due and immutable.

Time / event	Gate or surface	Operator action
After V1	V2 guide/version	Open at most one immutable V2.
10 calendar days after section V1 deadline	Version 2	Close; latest eligible version is grading candidate.

For a verified outage, use V1 +24 hours after service recovery. For a verified monthly cap, V1 is due 24 hours after the first accessible build day following the displayed reset; if no grant appears, extend through the first observed post-reset grant. V2 closes 10 calendar days after that personal V1 deadline. Record category/timing only - never billing screenshots, credentials, or account details.

4.5 Grading and review flow

- The 12-point checkpoint is formative only: User/value 2, Scope/state/interactions 3, Truth boundaries 2, Failure/access 2, Acceptance/verification 3. Score does not block building.
- Final artifact dimensions are Functionality, Craft, Relevance, and Verification evidence, each 0-10; totals are recomputed server-side.
- The provisional evaluator distinguishes submitted claim, observed behavior, and inference; it returns two citations, one observable gap, confidence, and approved flags per dimension.
- Deterministic policy caps dead/blocked links, absent AT/evidence, missing creator-to-public loop, unlabelled mocks, copied golden work without owned decisions, and generic work without industry/company transfer.
- Missing non-affiliation, benchmark-brand concern, or privacy/security evidence holds finalisation; privacy/security stops automated grading and suppresses the gallery.
- Low confidence, flags, outliers, appeals, dead links, mock ambiguity, and affiliation ambiguity enter instructor review.
- V2 is the latest eligible candidate even when it regresses; surface the regression rather than silently substituting V1.

4.6 Failure fallbacks

Trigger	Classroom response	Timing / evidence
Lovable outage for 10 min / majority	Freeze clock; continue contract, prompt, plan, AT mapping; use captioned recording/starter and failure hunt.	V1 due +24h after recovery; V2 +10 days from exception V1.
Daily credits exhausted below monthly cap	Complete plan/evidence and peer verification; use next daily grant for one bounded V2 request.	Ordinary V1/V2 window unless exception approved.

Trigger	Classroom response	Timing / evidence
Monthly cap reached	Do not promise next-day grant or require payment. Use validated zero-credit starter only if proven; otherwise plan + peer-verifier and <code>tool-access</code> .	Personal V1 after accessible reset day +24h; V2 +10 days.
Account/login failure	Try supported browser/profile and institution email; after 5 min use plan + peer-verifier.	Record <code>tool-access</code> ; never share login.
Section network failure	Use local deck/handout/recording, contract, prompt, plan, and paper AT design; test local fallback if provisioned.	Deferred publication rule.
Unsafe/expanded model plan	Do not approve; directly remove auth, payment, real integrations, secrets, and unrelated scope.	Record calibration observation.
Public URL/route fails	Republish changes, test base/public in incognito, use frozen base-origin fragment route only.	Submit honest failure if deadline; repair in V2.
Real/private data published	Unpublish, do not reproduce value, rotate any secret, remove data, republish, rerun privacy/incognito.	Privacy/security hold until cleared.

Accessible equivalents include tagged/text materials, captions/transcript, persistent labels/errors, visible focus, keyboard order, move buttons, no drag-only/color-only action, WCAG AA contrast, useful alt text, reduced motion, and evidence by timestamped screenshots + AT log, screen-reader transcript, peer witness, or transcribed audio.

4.7 Privacy, rights, and public truth

- Use fictional data, reserved demo identities, and original naming/copy/design.
- Show the disclosure: "Independent educational build. Not affiliated with or endorsed by the benchmark product."
- Label browser-local analytics and simulated integrations. Do not fabricate geography, connected services, or production parity.
- No auth, payment, real mailing, private API, real company/customer data, or credential is required for the core.
- Gallery uses only the latest finalised eligible version with consent, safe screenshot, caption, and link. It excludes grades, AI rationale/confidence/flags, first prompt/plan, TrustMRR evidence, and revenue claims.

4.8 Post-class operations

- ☐ Confirm V1 is immutable and V2 deadline is correct for the section or personal exception.

- ☐ Review low-confidence, policy/injection/privacy, link, outlier, mock, affiliation, and appeal cases.
- ☐ Complete the V2 backlog: one failed test and one craft/access repair; rerun regressions and republish.
- ☐ Add only final public URL, one-line user/job statement, verification evidence, and non-affiliation disclosure to the portfolio.
- ☐ Do not publish grade, grader rationale, prompt history, private TrustMRR data, or a "full clone" claim.
- ☐ Double-score calibration samples on the authored cadence and log observed Lovable UI/credit changes without changing cohort expectations silently.

Source binding: *session-04/01-lesson-plan.md, 13-instructor-fallback-plan.md, source-ledger.md, 10-rubric-grader-anchors.md, 11-accessibility-and-fallbacks.md, 09-lms-manifest.yaml, README.md.*

5. Session 05 - Revenue systems with Make

5.1 Promise, ownership, and boundaries

Student proof: one individually owned workflow design and prototype linked to the team/product context. Each learner submits their own revised flowchart, blueprint, five-case trace, output, workflow PNG, and ownership evidence. The later company-facing rollup is separate: usefulness/execution and verified sign-off may be team-level, while ownership stays individual.

Control model: explicit state; dedupe before irreversible work; invalid data quarantined; temporary failure receives bounded recovery; risky action waits for a recorded decision; every outcome leaves inspectable evidence.

Do not reward module count, animation, AI calls, or app logos. Reward complete control and verified business behavior.

5.2 Exact 120-minute run of show

Time	Min	Mode	Instructor move	Learner work / evidence
00:00-00:05	5	Arrival retrieval	Scramble five event cards; ask what must never happen twice.	Order events; name one irreversible action.
00:05-00:12	7	Consequential hook	Run the flawed scenario twice; show duplicate rows/drafts; ask what failed.	Predict consequence; name missing control.
00:12-00:20	8	Concept + proof	Teach trigger, contract, state, decision, action, evidence; demonstrate key/approval.	Mark the six boxes on template.

Time	Min	Mode	Instructor move	Learner work / evidence
00:20-00:35	15	Build 1: flowchart	Release one workflow pack; do not solve branches aloud.	Create initial chart + failure table with start, success, three non-happy outcomes.
00:35-00:42	7	Gate 1 feedback	Show feedback shape; diagnostic score is not a grade.	Submit problem frame + chart; flag incorrect advice.
00:42-00:52	10	Repair	Confer on unsafe actions/unbounded loops.	Revise chart; disposition each accepted/rejected suggestion.
00:52-01:02	10	Instructor live build	Build receive -> normalize/validate -> dedupe -> route -> queue/draft -> audit; normal only.	Map modules to flowchart nodes; note divergences.
01:02-01:27	25	Build 2: guided scenario	Checkpoints at minutes 70, 78, 87; protect individual ownership.	Build/import own copy, own connections, change one control, schedule off, pass normal.
01:27-01:41	14	Failure gauntlet	Release duplicate, malformed, timeout, approval after predictions; one repair.	Predict, run/replay, record actual; keep risky action queued/draft.
01:41-01:51	10	Evidence packaging	Demonstrate export, redaction, blueprint vs dynamic link.	Export JSON, redacted log/output, wide PNG, limitation/change note.
01:51-01:57	6	Gate 2 final submission	Run privacy checklist; preview gallery control.	Submit revised chart, blueprint, log, output, PNG, optional link, notes.
01:57-02:00	3	Reflection / bridge	Ask what would prevent real-company activation.	Write one refusal condition and one next test.

Protected student work: about 87 minutes. Preserve initial design, revision, normal/duplicate/approval, malformed/timeout replay, and evidence packaging. Use the offline replay to compress non-live traces if needed.

If the optional counted quiz is activated, run it at 00:00-00:07, omit retrieval, compress the duplicate hook to 00:07-00:12, and resume at minute 12. Do not take time from assessed build blocks.

5.3 Hook, build, and reveal control

Hook sequence: run `normal.json`; rerun unchanged; show two duplicate business actions; take three hypotheses; then name idempotency. Never use a real email or actually send a message - create a draft or `pending_approval` row.

Preferred build decisions:

1. Preserve `trace_id` on receive.
2. Normalize email and compute the idempotency key before action.
3. Validate required fields and consent.
4. Search the audit store for the key.

5. Route malformed, duplicate, approval-required, and ordinary cases.
6. Draft or queue; do not send.
7. Write final state, reason, owner, and timestamps.
8. Attach bounded Retry/incomplete-execution behavior only to a simulated transient failure.

Before Gate 1, do not release the corrected flowchart, expected-results manifest, duplicate-key expression, exact timeout recovery, or final quiz answers. The normal fixture/contract may be visible; reveal the other expected outcomes only after prediction.

5.4 Gates and submission timing

Gate / time	Condition	Operator action
S5.1 / session start to 00:42	Initial flowchart fields validate	Open design; close at minute 42; preserve version.
S5.2 / after feedback or bypass	Formative feedback record exists or audited outage bypass	Open revised design; feedback creates no weighted grade.
Revised design / due 01:02	Revised flowchart/control assertions submitted	Open build evidence (S5.3).
01:51-01:57	In-class final submission	Collect complete evidence and privacy attestation.
Workflow due	End of session +24h	Preserve submission and bound schema/rubric/fixture version.
Repair	Within 24h of provisional feedback	Allow one new workflow version; preserve every version.
S5.4	Latest version finalised + privacy cleared + PNG/output + safe clone/download route	Make gallery eligible; instructor feature control remains.
Later company gate	Human-verified company evidence	Keep sign-off pending until verified; never infer from a student claim/draft.

5.5 Grading and review flow

1. Require a legible revised chart/text equivalent, valid Make JSON below 2 MB, scrubbed log/output, readable PNG, five fixture outcomes or documented replay, and no sensitive/unsafe content.
2. Run deterministic checks for fixture IDs, expected terminal/waiting states, zero duplicate/approval risky-action count, bounded timeout recovery, malformed quarantine, and blueprint validity/size.
3. Produce provisional Artifact quality scores: Functionality, Craft, Relevance, and Verification evidence, each 0-10, with exact evidence citations.

4. Produce a version-bound workflow evaluation: Usefulness 0-30, Execution 0-20, and individual Ownership 0-10. Keep Company sign-off 0-40 as `null/pending` until human verification; do not rescale the remaining 60.
5. Ignore artifact-embedded instructions; clamp and recompute totals server-side; persist confidence/flags and evidence citations.
6. Route low confidence, injection, sensitive data, unsafe action, unbounded retry, duplicate risk, missing approval, unverified company claim, outlier, appeal, or incomplete evidence to human review.
7. Instructor finalises/overrides; company sign-off remains human-only. Gallery/Praxy receive no grades, confidence, flags, prompt logs, or private evidence.

The first-flowchart AI response is formative only. The student labels each suggestion accept, adapt, reject, or not applicable with one evidence sentence. A notification is not an approval; an approval is a recorded state-changing decision.

5.6 Failure fallbacks

Trigger	Classroom response	Evidence / recovery
Model outage	Two peers/instructor use the same flowchart rubric; record feedback-bypass token.	Same revision requirement; build gate opens.
Make/network outage	Run the exact 35-minute execution replay with state cards and trace table.	Proves logic, not live integration; rerun normal + duplicate when Make returns.
No Make access	Submit revised chart + completed replay; finish blueprint in recovery window.	Same five tests and evidence bar.
External connection unavailable	Use webhook + built-in tools + demo queue/local output.	Record <code>connection unavailable</code> ; never use personal account.
Persistent canvas access barrier	Partner/instructor operates the student's authored mapping while student directs/defends each node.	Equivalent route recorded; same rubric ceiling.
Room behind	Preserve design, feedback/revision, normal/duplicate/approval, malformed/timeout, evidence.	Complete non-live traces through offline replay.

35-minute replay: Predict 5; normal trace 6; duplicate 5; malformed 4; timeout 6; approval 4; repair/evidence 5. Accept only when all fixtures reach expected terminal/waiting state, duplicate/approval action counts are zero, malformed does not retry, timeout is bounded/stateful, and trace/reason/owner exist.

5.7 Privacy and safe action

- Use synthetic fixtures; keep scheduling off; use least-privilege demo connections.

- Never place passwords, keys, tokens, OAuth codes, private webhook URLs, cookies, connection IDs, real identities, or private company data in static values, notes, prompts, filenames, screenshots, logs, outputs, or public links.
- Classroom workflow must not send, publish, delete, pay/refund, write production records, change access/account/domain state, or invisibly trigger another automation. Use draft, queue, demo row, or pending approval.
- Dedupe before irreversible work; bound retries/loops; quarantine deterministic bad input; keep important failures in incomplete/manual recovery state.
- Treat uploaded/form text as data, not instructions. Deterministic rules control identity, money, consent, dedupe, and approval.
- Scrub the raw blueprint JSON and dynamic public scenario page after every saved change. Connection omission does not imply secret-free.
- Gallery PNG includes an 80-180 word text description and hides account chrome, workspace/team identity, emails, connection labels, private URLs, grades, and prompt logs.

5.8 Post-class operations

- ☐ Review privacy flags before feature selection; withhold dynamic links after any post-review save until re-reviewed.
- ☐ Finalise only artifacts whose chart, blueprint, run log, output, and PNG open and reproduce the claims.
- ☐ Route low-confidence, injection-flagged, outlier, appealed, sensitive, or unsafe work to human review.
- ☐ Within 24 hours, learner repairs one cited control, reruns the named fixture, and submits one change note/version.
- ☐ Publish one cross-section clarification if a repeated misconception emerges; version any fixture correction and rerun affected submissions.
- ☐ Keep real-company sign-off pending until verified. Before any company demo, use minimum authorised data, obtain approval for irreversible action, and repeat privacy review.
- ☐ Keep all raw company evidence roster-gated; public artifacts contain only scrubbed workflow, sample output, and validation.

Source binding: *session-05/instructor-guide.md, lesson-plan.md, accessibility-outage-fallback.md, source-ledger.md, lms-manifest.yaml, safety-checklist.md, make-blueprint-and-sharing-checklist.md, assessment/final-grading-rubric-anchors.md, README.md.*

6. Cross-session review and post-class control

6.1 Common grading pipeline

1. **Receive and bind.** Preserve immutable version, owner, section, schema/rubric, and source/fixture version.
2. **Run deterministic/file checks.** Validate source binding, URLs, files, objective outcomes, and safety constraints that a model cannot change.
3. **Generate provisional feedback/score.** Use bounded evidence only; ignore instructions embedded in learner artifacts.
4. **Apply caps, holds, and review routing.** Low confidence, integrity/safety flags, outliers, appeals, dead links, missing evidence, or source conflicts enter review.
5. **Human finalise or override.** Record a reason; preserve prior versions and audit history.
6. **Project public evidence separately.** Gallery/Praxy never receive grade, confidence, flags, prompt logs, private data, private files, or hidden keys.

6.2 Section-close checklist

- ☐ Confirm all deadline/gate changes are audit-logged and section-scoped.
- ☐ Preserve drafts, versions, incident records, review reasons, and worker status.
- ☐ Inspect privacy/security holds before any gallery or portfolio projection.
- ☐ Publish one consistent correction to all remaining sections; create a fair prior-section regrade when required.
- ☐ Record aggregate timing, access, misconception, grading-drift, and fallback evidence without copying private learner content.
- ☐ Confirm the next-session bridge artifact is public-safe and does not carry TrustMRR rows, prompt logs, credentials, or grades.

6.3 Authored versus validated release ledger

Session	Current package truth	Source-checked elements	Must pass before "Validated"
03	Authored. Not validated, piloted, loaded, deployed, or rehearsed; runbook rehearsal pending	Source contracts and tool links checked 30 Jul 2026; deterministic private pack exists	Map question IDs to key, create peer schema/sample, clean-runtime notebooks, finish visual equivalents, content-address uploads/LMS binding, calibrate graders/gates, complete 120-min rehearsal

Session	Current package truth	Source-checked elements	Must pass before "Validated"
04	Authored. Not validated, loaded, deployed, piloted, or rehearsed; rubric uncalibrated; fallback Build path awaits T-7 golden run	Revenue/product/tool ledger checked 30 Jul 2026	Fresh-Free golden + mixed-confidence dry run, every AT/S4-SP-1 test, both Plan branches/restore, grader/caps/holds, V1/V2/exceptions, captioned fallback assets, T-7 recheck, dummy-only public examples
05	Authored. Not validated, piloted, loaded, deployed, or rehearsed	Official Make ledger checked 30 Jul 2026; fixture contracts authored	T-7 claims, 60-person feedback/bypass load test, actual five-fixture run with redacted evidence, 12+ grader calibration examples, staged LMS fields/gates/gallery actions, privacy review of blueprint/link, live + outage rehearsal

No-Go wording: Do not call a source-checked claim, a generated fixture pack, a successful demo, or an authored fallback "session validation." Advance lifecycle only through the course release gate.

7. Source recheck ledger

Record the reviewer, date, result, and affected files. If a volatile claim changes, update lesson, deck, handout, LMS copy, rubric/quiz where affected, and this manual together.

7.1 Session 03 ledger

Claim/control	Source	Package check	Recheck / operator record
Course arc and verified-data-memo outcome	lms/docs/build/04_course_outline_COT_v3.md	30 Jul 2026	On course-contract change: _____
Mixed grading and over-context reveal	lms/docs/build/10_sessions_3_5_redesign_brief.md	30 Jul 2026	Before production lock: _____
Artifact rubric and privacy separation	lms/docs/build/01_scoring_methodology.md	30 Jul 2026	On scoring-method change: _____
Delivered Sheets/upload bridge and pacing	lms/docs/taught/2026-07-workshop-sessions-01-02-reference.pdf	30 Jul 2026	Delivered evidence; no recurring recheck
Google Sheets FILTER/MEDIAN/QUERY behavior	Google Sheets function list	30 Jul 2026	T-7: reviewer/date/result _____
Colab runtime, storage, and sharing cautions	Google Colab FAQ	30 Jul 2026	T-7: reviewer/date/result _____
pandas CSV//IO behavior	pandas I/O reference	30 Jul 2026	T-7: reviewer/date/result _____

7.2 Session 04 ledger

Claim/control	Source	Classroom treatment	T-7 operator record
Current Liinks revenue credibility	TrustMRR Liinks profile ; founder site	Use rounded "about \$30K MRR" only after current check; never imply fixed	_____
HabitKit alternate credibility	TrustMRR HabitKit ; Google Play	Selection comparison only	_____
QR Code AI alternate credibility	TrustMRR QR Code AI ; official site	Selection comparison only	_____
Liinks create/edit/reorder/design behavior	Create a page	Core benchmark behavior; original brand/design	_____
Block catalogue	Block types	Six representative types; label omitted parity	_____
Analytics surface	Liinks analytics	Browser-local demo only; no fabricated geography	_____
Plan/API/domain boundaries	Plans ; API ; Domains	No copied pricing/billing, API, or fake DNS	_____
Lovable Free credits and Build variability	Pricing ; Credits and usage	No credit promise or purchase requirement	_____
Plan mode behavior/cost and no-formal-plan branch	Plan mode	One Plan message; edit formal plan or use gated fallback	_____
Publish cost, public visibility, and republish requirement	Publish	Use dialog; dummy data; incognito; republish V2	_____

7.3 Session 05 ledger

Claim/control	Official source	Classroom consequence	T-7 operator record
Blueprint content, import, connections, and <2 MB limit	Make blueprints	Export point-in-time JSON; importer recreates connections	_____
Public scenario visibility, copy, latest-saved behavior, omissions	Scenario sharing	Logged-out scrub; dynamic link withdrawn after unreviewed save	_____
Router order and fallback route	Make router	Sequential routes; explicit fallback for unmatched data	_____
Skip/Retry/Resume/Commit/Rollback semantics	Error handling	Retry temporary faults; never silently skip high-impact work	_____

Claim/control	Official source	Classroom consequence	T-7 operator record
Incomplete execution, ordering, retention, commit, consecutive errors	Scenario settings	Settings are operational controls; document current behavior	_____
Instant webhook parallelism, queues, and responses	Webhooks	Use ordering only when needed; discuss concurrency/idempotency	_____
History fields and plan-dependent retention/search	Scenario history	Capture minimum redacted proof; do not assume indefinite history	_____
Retry/manual resolution for temporary vs configuration/data faults	Manage incomplete executions	Bounded retry, then manual owner; malformed input quarantines	_____
Free credits, active-scenario cap, interval, run time, file limit	Make pricing	Keep inactive/manual fixture runs; do not rely on paid-only limits/features	_____

8. Instructor handoff note

Before handing a section to another instructor, give them: source-package version, source-recheck record, section deadline/exception rule, gate state, demo/fallback location, known tool drift, active incident, reviewer/finaliser, and any cohort-wide correction. Do not hand off private keys, credentials, or learner data in chat or an unapproved shared drive.